Assignment 1:

Name: Leon Lobo

Admission No: U24AI043

Q1. Consider a map given in image

Implement BFS and DFS algorithm to find all possible step costs between Syracuse and Chicago.

Code:
```
from collections import deque

def bfs(graph, start, end):
    q = deque()
    q.append((start, 0, [start], {start}))

    results = []

    while q:
```

```
        node, cost, path, visited = q.popleft()

        if node == end:
            results.append((path, cost))
            continue

        for neighbour, weight in graph[node]:
            if neighbour not in visited:
                q.append((
                    neighbour,
                    cost + weight,
                    path + [neighbour],        # new path
                    visited | {neighbour}      # new visited set
                ))

    return results


def dfs(graph, start, end):
    results = []

    def df_search(node, cost, path, visited):
        if node == end:
            results.append((path[:], cost))
            return

        for neighbour, weight in graph[node]:
            if neighbour not in visited:
                df_search(
                    neighbour,
                    cost + weight,
                    path + [neighbour],       # new path
                    visited | {neighbour}     # new visited set
                )

    df_search(start, 0, [start], {start})
    return results




def main():
    graph = {
    "Syracuse": [("Buffalo",150),("Boston",312),("New York",254),("Philadelphia",253)],
    "Buffalo": [("Syracuse",150),("Detroit",256),("Cleveland",189),("Pittsburgh",215)],
    "Detroit": [("Buffalo",256),("Chicago",283)],
    "Cleveland":
[("Buffalo",189),("Chicago",345),("Detroit",169),("Columbus",144),("Pittsburgh",134)],
```

```
    "Pittsburgh":
[("Buffalo",215),("Cleveland",134),("Columbus",185),("Baltimore",247),("Philadelphia",305)],
    "Columbus": [("Pittsburgh",185),("Indianapolis",176),("Cleveland",144)],
    "Indianapolis": [("Columbus",176),("Chicago",182)],
    "New York": [("Syracuse",254),("Philadelphia",97),("Providence",181),("Boston",215)],
    "Philadelphia": [("New York",97),("Baltimore",101),("Pittsburgh",305),("Syracuse",253)],
    "Baltimore": [("Philadelphia",101),("Pittsburgh",247)],
    "Providence": [("New York",181),("Boston",50)],
    "Boston": [("Syracuse",312),("Providence",50),("Portland",107)],
    "Portland": [("Boston",107)],
    "Chicago": []}
    algo=int(input("Enter 1 for BFS, 2 for DFS:"))
    start=input("Enter starting city:").strip().capitalize()
    end=input("Enter end city:").strip().capitalize()
    if algo==1:
        bfs_costs=bfs(graph,start,end)
        for path, cost in bfs_costs:
            for city in path:
                print(city,"-",end=" ")
            print(cost)
    else:
        dfs_costs=dfs(graph,start,end)
        for path, cost in dfs_costs:
            for city in path:
                print(city,"-",end=" ")
            print(cost)




if __name__=='__main__':
    main()
```

Ouput: BFS-

```
PS C:\Education\SVNIT\Coding\AI> python -u "c:\Education\SVNIT\Coding\AI\Lab 1\A1_Q1.py"
Enter 1 for BFS, 2 for DFS:1
Enter starting city:Syracuse
Enter end city:Chicago
Syracuse - Buffalo - Detroit - Chicago - 689
Syracuse - Buffalo - Cleveland - Chicago - 684
Syracuse - Buffalo - Cleveland - Detroit - Chicago - 791
Syracuse - Buffalo - Pittsburgh - Cleveland - Chicago - 844
Syracuse - Philadelphia - Pittsburgh - Cleveland - Chicago - 1037
Syracuse - Buffalo - Cleveland - Columbus - Indianapolis - Chicago - 841
Syracuse - Buffalo - Pittsburgh - Cleveland - Detroit - Chicago - 951
Syracuse - Buffalo - Pittsburgh - Columbus - Indianapolis - Chicago - 908
Syracuse - Buffalo - Pittsburgh - Columbus - Cleveland - Chicago - 1039
Syracuse - New York - Philadelphia - Pittsburgh - Cleveland - Chicago - 1135
Syracuse - Philadelphia - Baltimore - Pittsburgh - Cleveland - Chicago - 1080
Syracuse - Philadelphia - Pittsburgh - Buffalo - Detroit - Chicago - 1312
Syracuse - Philadelphia - Pittsburgh - Buffalo - Cleveland - Chicago - 1307
Syracuse - Philadelphia - Pittsburgh - Cleveland - Detroit - Chicago - 1144
Syracuse - Philadelphia - Pittsburgh - Columbus - Indianapolis - Chicago - 1101
Syracuse - Philadelphia - Pittsburgh - Columbus - Cleveland - Chicago - 1232
Syracuse - Buffalo - Cleveland - Pittsburgh - Columbus - Indianapolis - Chicago - 1016
Syracuse - Buffalo - Pittsburgh - Cleveland - Columbus - Indianapolis - Chicago - 1001
Syracuse - Buffalo - Pittsburgh - Columbus - Cleveland - Detroit - Chicago - 1146
Syracuse - New York - Philadelphia - Baltimore - Pittsburgh - Cleveland - Chicago - 1178
Syracuse - New York - Philadelphia - Pittsburgh - Buffalo - Detroit - Chicago - 1410
Syracuse - New York - Philadelphia - Pittsburgh - Buffalo - Cleveland - Chicago - 1405
Syracuse - New York - Philadelphia - Pittsburgh - Cleveland - Detroit - Chicago - 1242
Syracuse - New York - Philadelphia - Pittsburgh - Columbus - Indianapolis - Chicago - 1199
Syracuse - New York - Philadelphia - Pittsburgh - Columbus - Cleveland - Chicago - 1330
Syracuse - Philadelphia - Baltimore - Pittsburgh - Buffalo - Detroit - Chicago - 1355
Syracuse - Philadelphia - Baltimore - Pittsburgh - Buffalo - Cleveland - Chicago - 1350
Syracuse - Philadelphia - Baltimore - Pittsburgh - Cleveland - Detroit - Chicago - 1187
Syracuse - Philadelphia - Baltimore - Pittsburgh - Columbus - Indianapolis - Chicago - 1144
Syracuse - Philadelphia - Baltimore - Pittsburgh - Columbus - Cleveland - Chicago - 1275
Syracuse - Philadelphia - Pittsburgh - Buffalo - Cleveland - Detroit - Chicago - 1414
Syracuse - Philadelphia - Pittsburgh - Cleveland - Buffalo - Detroit - Chicago - 1420
Syracuse - Philadelphia - Pittsburgh - Cleveland - Columbus - Indianapolis - Chicago - 1194
Syracuse - Philadelphia - Pittsburgh - Columbus - Cleveland - Detroit - Chicago - 1339
Syracuse - Boston - Providence - New York - Philadelphia - Pittsburgh - Cleveland - Chicago - 1424
Syracuse - New York - Philadelphia - Baltimore - Pittsburgh - Buffalo - Detroit - Chicago - 1453
```

```
Syracuse - New York - Philadelphia - Baltimore - Pittsburgh - Buffalo - Cleveland - Chicago - 1448
Syracuse - New York - Philadelphia - Baltimore - Pittsburgh - Cleveland - Detroit - Chicago - 1285
Syracuse - New York - Philadelphia - Baltimore - Pittsburgh - Columbus - Indianapolis - Chicago - 1242
Syracuse - New York - Philadelphia - Baltimore - Pittsburgh - Columbus - Cleveland - Chicago - 1373
Syracuse - New York - Philadelphia - Pittsburgh - Buffalo - Cleveland - Detroit - Chicago - 1512
Syracuse - New York - Philadelphia - Pittsburgh - Cleveland - Buffalo - Detroit - Chicago - 1518
Syracuse - New York - Philadelphia - Pittsburgh - Cleveland - Columbus - Indianapolis - Chicago - 1292
Syracuse - New York - Philadelphia - Pittsburgh - Columbus - Cleveland - Detroit - Chicago - 1437
Syracuse - Philadelphia - Baltimore - Pittsburgh - Buffalo - Cleveland - Detroit - Chicago - 1457
Syracuse - Philadelphia - Baltimore - Pittsburgh - Cleveland - Buffalo - Detroit - Chicago - 1463
Syracuse - Philadelphia - Baltimore - Pittsburgh - Cleveland - Columbus - Indianapolis - Chicago - 1237
Syracuse - Philadelphia - Baltimore - Pittsburgh - Columbus - Cleveland - Detroit - Chicago - 1382
Syracuse - Philadelphia - Pittsburgh - Buffalo - Cleveland - Columbus - Indianapolis - Chicago - 1464
Syracuse - Philadelphia - Pittsburgh - Columbus - Cleveland - Buffalo - Detroit - Chicago - 1615
Syracuse - Boston - Providence - New York - Philadelphia - Baltimore - Pittsburgh - Cleveland - Chicago - 1467
Syracuse - Boston - Providence - New York - Philadelphia - Pittsburgh - Buffalo - Detroit - Chicago - 1699
Syracuse - Boston - Providence - New York - Philadelphia - Pittsburgh - Buffalo - Cleveland - Chicago - 1694
Syracuse - Boston - Providence - New York - Philadelphia - Pittsburgh - Cleveland - Detroit - Chicago - 1531
Syracuse - Boston - Providence - New York - Philadelphia - Pittsburgh - Columbus - Indianapolis - Chicago - 1488
Syracuse - Boston - Providence - New York - Philadelphia - Pittsburgh - Columbus - Cleveland - Chicago - 1619
Syracuse - New York - Philadelphia - Baltimore - Pittsburgh - Buffalo - Cleveland - Detroit - Chicago - 1555
Syracuse - New York - Philadelphia - Baltimore - Pittsburgh - Cleveland - Buffalo - Detroit - Chicago - 1561
Syracuse - New York - Philadelphia - Baltimore - Pittsburgh - Cleveland - Columbus - Indianapolis - Chicago - 1335
Syracuse - New York - Philadelphia - Baltimore - Pittsburgh - Columbus - Cleveland - Detroit - Chicago - 1480
Syracuse - New York - Philadelphia - Pittsburgh - Buffalo - Cleveland - Columbus - Indianapolis - Chicago - 1562
Syracuse - New York - Philadelphia - Pittsburgh - Columbus - Cleveland - Buffalo - Detroit - Chicago - 1713
Syracuse - Philadelphia - Baltimore - Pittsburgh - Buffalo - Cleveland - Columbus - Indianapolis - Chicago - 1507
Syracuse - Philadelphia - Baltimore - Pittsburgh - Columbus - Cleveland - Buffalo - Detroit - Chicago - 1658
Syracuse - Boston - Providence - New York - Philadelphia - Baltimore - Pittsburgh - Buffalo - Detroit - Chicago - 1742
Syracuse - Boston - Providence - New York - Philadelphia - Baltimore - Pittsburgh - Buffalo - Cleveland - Chicago - 1737
Syracuse - Boston - Providence - New York - Philadelphia - Baltimore - Pittsburgh - Cleveland - Detroit - Chicago - 1574
Syracuse - Boston - Providence - New York - Philadelphia - Baltimore - Pittsburgh - Columbus - Indianapolis - Chicago - 1531
Syracuse - Boston - Providence - New York - Philadelphia - Baltimore - Pittsburgh - Columbus - Cleveland - Chicago - 1662
Syracuse - Boston - Providence - New York - Philadelphia - Pittsburgh - Buffalo - Cleveland - Detroit - Chicago - 1801
Syracuse - Boston - Providence - New York - Philadelphia - Pittsburgh - Cleveland - Buffalo - Detroit - Chicago - 1807
Syracuse - Boston - Providence - New York - Philadelphia - Pittsburgh - Cleveland - Columbus - Indianapolis - Chicago - 1581
Syracuse - Boston - Providence - New York - Philadelphia - Pittsburgh - Columbus - Cleveland - Detroit - Chicago - 1726
Syracuse - New York - Philadelphia - Baltimore - Pittsburgh - Buffalo - Cleveland - Columbus - Indianapolis - Chicago - 1605
Syracuse - New York - Philadelphia - Baltimore - Pittsburgh - Columbus - Cleveland - Buffalo - Detroit - Chicago - 1756
Syracuse - Boston - Providence - New York - Philadelphia - Baltimore - Pittsburgh - Buffalo - Cleveland - Detroit - Chicago - 1844
```

```
Syracuse - Boston - Providence - New York - Philadelphia - Baltimore - Pittsburgh - Cleveland - Buffalo - Detroit - Chicago - 1850
Syracuse - Boston - Providence - New York - Philadelphia - Baltimore - Pittsburgh - Cleveland - Columbus - Indianapolis - Chicago - 1624
Syracuse - Boston - Providence - New York - Philadelphia - Baltimore - Pittsburgh - Columbus - Cleveland - Detroit - Chicago - 1769
Syracuse - Boston - Providence - New York - Philadelphia - Pittsburgh - Buffalo - Cleveland - Columbus - Indianapolis - Chicago - 1851
Syracuse - Boston - Providence - New York - Philadelphia - Pittsburgh - Columbus - Cleveland - Buffalo - Detroit - Chicago - 2002
Syracuse - Boston - Providence - New York - Philadelphia - Baltimore - Pittsburgh - Buffalo - Cleveland - Columbus - Indianapolis - Chicago - 1894
```

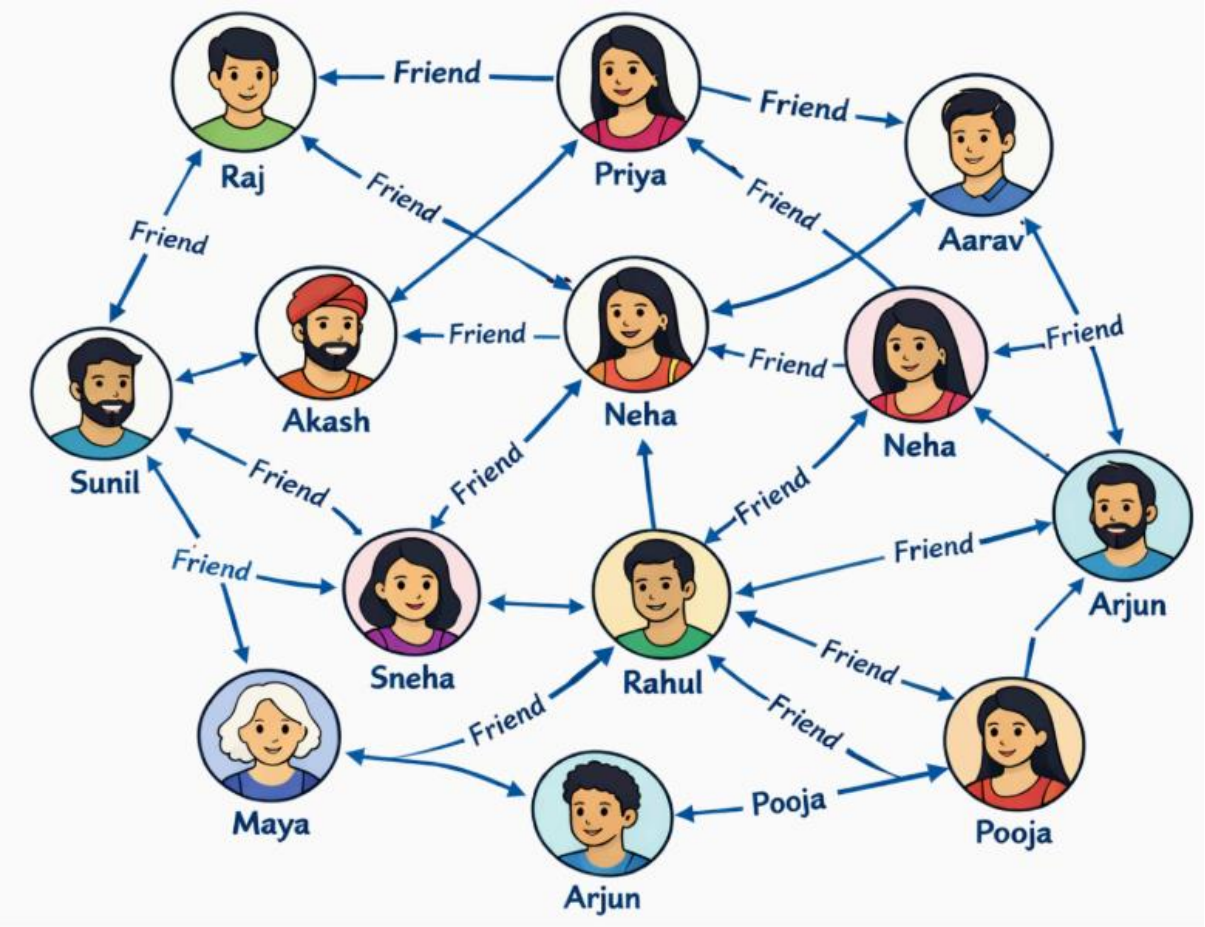
DFS-

```
PS C:\Education\SVNIT\Coding\AI> python -u "c:\Education\SVNIT\Coding\AI\Lab 1\A1_Q1.py"
Enter 1 for BFS, 2 for DFS:2
Enter starting city:SYRACUSE
Enter end city:chicago
Syracuse - Buffalo - Detroit - Chicago - 689
Syracuse - Buffalo - Cleveland - Chicago - 684
Syracuse - Buffalo - Cleveland - Detroit - Chicago - 791
Syracuse - Buffalo - Cleveland - Columbus - Indianapolis - Chicago - 841
Syracuse - Buffalo - Cleveland - Pittsburgh - Columbus - Indianapolis - Chicago - 1016
Syracuse - Buffalo - Pittsburgh - Cleveland - Chicago - 844
Syracuse - Buffalo - Pittsburgh - Cleveland - Detroit - Chicago - 951
Syracuse - Buffalo - Pittsburgh - Cleveland - Columbus - Indianapolis - Chicago - 1001
Syracuse - Buffalo - Pittsburgh - Columbus - Indianapolis - Chicago - 908
Syracuse - Buffalo - Pittsburgh - Columbus - Cleveland - Chicago - 1039
Syracuse - Buffalo - Pittsburgh - Columbus - Cleveland - Detroit - Chicago - 1146
Syracuse - Boston - Providence - New York - Philadelphia - Baltimore - Pittsburgh - Buffalo - Detroit - Chicago - 1742
Syracuse - Boston - Providence - New York - Philadelphia - Baltimore - Pittsburgh - Buffalo - Cleveland - Chicago - 1737
Syracuse - Boston - Providence - New York - Philadelphia - Baltimore - Pittsburgh - Buffalo - Cleveland - Detroit - Chicago - 1844
Syracuse - Boston - Providence - New York - Philadelphia - Baltimore - Pittsburgh - Buffalo - Cleveland - Columbus - Indianapolis - Chicago - 1894
Syracuse - Boston - Providence - New York - Philadelphia - Baltimore - Pittsburgh - Cleveland - Buffalo - Detroit - Chicago - 1850
Syracuse - Boston - Providence - New York - Philadelphia - Baltimore - Pittsburgh - Cleveland - Chicago - 1467
Syracuse - Boston - Providence - New York - Philadelphia - Baltimore - Pittsburgh - Cleveland - Detroit - Chicago - 1574
Syracuse - Boston - Providence - New York - Philadelphia - Baltimore - Pittsburgh - Cleveland - Columbus - Indianapolis - Chicago - 1624
Syracuse - Boston - Providence - New York - Philadelphia - Baltimore - Pittsburgh - Columbus - Indianapolis - Chicago - 1531
Syracuse - Boston - Providence - New York - Philadelphia - Baltimore - Pittsburgh - Columbus - Cleveland - Buffalo - Detroit - Chicago - 2045
Syracuse - Boston - Providence - New York - Philadelphia - Baltimore - Pittsburgh - Columbus - Cleveland - Chicago - 1662
Syracuse - Boston - Providence - New York - Philadelphia - Baltimore - Pittsburgh - Columbus - Cleveland - Detroit - Chicago - 1769
Syracuse - Boston - Providence - New York - Philadelphia - Pittsburgh - Buffalo - Detroit - Chicago - 1699
Syracuse - Boston - Providence - New York - Philadelphia - Pittsburgh - Buffalo - Cleveland - Chicago - 1694
Syracuse - Boston - Providence - New York - Philadelphia - Pittsburgh - Buffalo - Cleveland - Detroit - Chicago - 1801
Syracuse - Boston - Providence - New York - Philadelphia - Pittsburgh - Buffalo - Cleveland - Columbus - Indianapolis - Chicago - 1851
Syracuse - Boston - Providence - New York - Philadelphia - Pittsburgh - Cleveland - Buffalo - Detroit - Chicago - 1807
Syracuse - Boston - Providence - New York - Philadelphia - Pittsburgh - Cleveland - Chicago - 1424
Syracuse - Boston - Providence - New York - Philadelphia - Pittsburgh - Cleveland - Detroit - Chicago - 1531
Syracuse - Boston - Providence - New York - Philadelphia - Pittsburgh - Cleveland - Columbus - Indianapolis - Chicago - 1581
Syracuse - Boston - Providence - New York - Philadelphia - Pittsburgh - Columbus - Indianapolis - Chicago - 1488
Syracuse - Boston - Providence - New York - Philadelphia - Pittsburgh - Columbus - Cleveland - Buffalo - Detroit - Chicago - 2002
Syracuse - Boston - Providence - New York - Philadelphia - Pittsburgh - Columbus - Cleveland - Chicago - 1619
Syracuse - Boston - Providence - New York - Philadelphia - Pittsburgh - Columbus - Cleveland - Detroit - Chicago - 1726
Syracuse - New York - Philadelphia - Baltimore - Pittsburgh - Buffalo - Detroit - Chicago - 1453
```

```
Syracuse - New York - Philadelphia - Baltimore - Pittsburgh - Buffalo - Cleveland - Chicago - 1448
Syracuse - New York - Philadelphia - Baltimore - Pittsburgh - Buffalo - Cleveland - Detroit - Chicago - 1555
Syracuse - New York - Philadelphia - Baltimore - Pittsburgh - Buffalo - Cleveland - Columbus - Indianapolis - Chicago - 1605
Syracuse - New York - Philadelphia - Baltimore - Pittsburgh - Cleveland - Buffalo - Detroit - Chicago - 1561
Syracuse - New York - Philadelphia - Baltimore - Pittsburgh - Cleveland - Chicago - 1178
Syracuse - New York - Philadelphia - Baltimore - Pittsburgh - Cleveland - Detroit - Chicago - 1285
Syracuse - New York - Philadelphia - Baltimore - Pittsburgh - Cleveland - Columbus - Indianapolis - Chicago - 1335
Syracuse - New York - Philadelphia - Baltimore - Pittsburgh - Columbus - Indianapolis - Chicago - 1242
Syracuse - New York - Philadelphia - Baltimore - Pittsburgh - Columbus - Cleveland - Buffalo - Detroit - Chicago - 1756
Syracuse - New York - Philadelphia - Baltimore - Pittsburgh - Columbus - Cleveland - Chicago - 1373
Syracuse - New York - Philadelphia - Baltimore - Pittsburgh - Columbus - Cleveland - Detroit - Chicago - 1480
Syracuse - New York - Philadelphia - Pittsburgh - Buffalo - Detroit - Chicago - 1410
Syracuse - New York - Philadelphia - Pittsburgh - Buffalo - Cleveland - Chicago - 1405
Syracuse - New York - Philadelphia - Pittsburgh - Buffalo - Cleveland - Detroit - Chicago - 1512
Syracuse - New York - Philadelphia - Pittsburgh - Buffalo - Cleveland - Columbus - Indianapolis - Chicago - 1562
Syracuse - New York - Philadelphia - Pittsburgh - Cleveland - Buffalo - Detroit - Chicago - 1518
Syracuse - New York - Philadelphia - Pittsburgh - Cleveland - Chicago - 1135
Syracuse - New York - Philadelphia - Pittsburgh - Cleveland - Detroit - Chicago - 1242
Syracuse - New York - Philadelphia - Pittsburgh - Cleveland - Columbus - Indianapolis - Chicago - 1292
Syracuse - New York - Philadelphia - Pittsburgh - Columbus - Indianapolis - Chicago - 1199
Syracuse - New York - Philadelphia - Pittsburgh - Columbus - Cleveland - Buffalo - Detroit - Chicago - 1713
Syracuse - New York - Philadelphia - Pittsburgh - Columbus - Cleveland - Chicago - 1330
Syracuse - New York - Philadelphia - Pittsburgh - Columbus - Cleveland - Detroit - Chicago - 1437
Syracuse - Philadelphia - Baltimore - Pittsburgh - Buffalo - Detroit - Chicago - 1355
Syracuse - Philadelphia - Baltimore - Pittsburgh - Buffalo - Cleveland - Chicago - 1350
Syracuse - Philadelphia - Baltimore - Pittsburgh - Buffalo - Cleveland - Detroit - Chicago - 1457
Syracuse - Philadelphia - Baltimore - Pittsburgh - Buffalo - Cleveland - Columbus - Indianapolis - Chicago - 1507
Syracuse - Philadelphia - Baltimore - Pittsburgh - Cleveland - Buffalo - Detroit - Chicago - 1463
Syracuse - Philadelphia - Baltimore - Pittsburgh - Cleveland - Chicago - 1080
Syracuse - Philadelphia - Baltimore - Pittsburgh - Cleveland - Detroit - Chicago - 1187
Syracuse - Philadelphia - Baltimore - Pittsburgh - Cleveland - Columbus - Indianapolis - Chicago - 1237
Syracuse - Philadelphia - Baltimore - Pittsburgh - Columbus - Indianapolis - Chicago - 1144
Syracuse - Philadelphia - Baltimore - Pittsburgh - Columbus - Cleveland - Buffalo - Detroit - Chicago - 1658
Syracuse - Philadelphia - Baltimore - Pittsburgh - Columbus - Cleveland - Chicago - 1275
Syracuse - Philadelphia - Baltimore - Pittsburgh - Columbus - Cleveland - Detroit - Chicago - 1382
Syracuse - Philadelphia - Pittsburgh - Buffalo - Detroit - Chicago - 1312
Syracuse - Philadelphia - Pittsburgh - Buffalo - Cleveland - Chicago - 1307
Syracuse - Philadelphia - Pittsburgh - Buffalo - Cleveland - Detroit - Chicago - 1414
Syracuse - Philadelphia - Pittsburgh - Buffalo - Cleveland - Columbus - Indianapolis - Chicago - 1464
Syracuse - Philadelphia - Pittsburgh - Cleveland - Buffalo - Detroit - Chicago - 1420
```

```
Syracuse - Philadelphia - Pittsburgh - Cleveland - Chicago - 1037
Syracuse - Philadelphia - Pittsburgh - Cleveland - Detroit - Chicago - 1144
Syracuse - Philadelphia - Pittsburgh - Cleveland - Columbus - Indianapolis - Chicago - 1194
Syracuse - Philadelphia - Pittsburgh - Columbus - Indianapolis - Chicago - 1101
Syracuse - Philadelphia - Pittsburgh - Columbus - Cleveland - Buffalo - Detroit - Chicago - 1615
Syracuse - Philadelphia - Pittsburgh - Columbus - Cleveland - Chicago - 1232
Syracuse - Philadelphia - Pittsburgh - Columbus - Cleveland - Detroit - Chicago - 1339
```

Q2. Consider a social media network as given below. Find BFS and DFS tree.



Code:

```
from collections import deque

def bfs_tree(graph, start):
    queue = deque([start])
    visited = {start}

    parent = {start: None}

    while queue:
        node = queue.popleft()

        for neighbour in graph.get(node, []):
            if neighbour not in visited:
                visited.add(neighbour)
                parent[neighbour] = node
                queue.append(neighbour)

    return parent

def dfs_tree(graph, start):
    visited = set()
    parent = {start: None}
```

```python
    def dfs(node):
        visited.add(node)
        for neighbour in graph.get(node, []):
            if neighbour not in visited:
                parent[neighbour] = node
                dfs(neighbour)

    dfs(start)
    return parent


def get_path(parent, node):
    path = []
    while node is not None:
        path.append(node)
        node = parent[node]
    return path[::-1]



def main():
    graph = {
        "Raj": ["Sunil", "Neha_1"],
        "Sunil": ["Raj", "Akash", "Sneha", "Maya"],
        "Akash": ["Sunil", "Priya"],
        "Priya": ["Raj", "Akash", "Aarav"],
        "Neha_1": ["Raj", "Akash", "Sneha", "Aarav"],
        "Sneha": ["Sunil", "Rahul", "Neha_1"],
        "Maya": ["Sunil", "Rahul", "Arjun_1"],
        "Rahul": ["Sneha", "Maya", "Pooja", "Arjun_2", "Neha_2", "Neha_1"],
        "Arjun_1": ["Maya", "Pooja"],
        "Pooja": ["Arjun_1", "Rahul", "Arjun_2"],
        "Arjun_2": ["Rahul", "Aarav", "Neha_2"],
        "Neha_2": ["Rahul", "Aarav", "Arjun_2", "Neha_1", "Priya"],
        "Aarav": ["Neha_1", "Neha_2", "Arjun_2"]
    }
    algo=int(input("Enter 1 for BFS, 2 for DFS:"))
    start=input("Enter start node:").strip().capitalize()
    if algo==1:
        parent = bfs_tree(graph, start)
        # Find all nodes that are parents
        parents = set(p for p in parent.values() if p is not None)

        # Leaf nodes = nodes that never act as a parent
        leaf_nodes = [node for node in parent if node not in parents]
        for leaf in leaf_nodes:
```

```
            path = get_path(parent,leaf)
            print(" -> ".join(path))
    else:
        parent = dfs_tree(graph, start)
        # Find all nodes that are parents
        parents = set(p for p in parent.values() if p is not None)

        # Leaf nodes = nodes that never act as a parent
        leaf_nodes = [node for node in parent if node not in parents]
        for leaf in leaf_nodes:
            path = get_path(parent,leaf)
            print(" -> ".join(path))





if __name__=='__main__':
    main()
```

Output:

```
Enter 1 for BFS, 2 for DFS:1
Enter start node:Rahul
Rahul -> Pooja
Rahul -> Sneha -> Sunil
Rahul -> Maya -> Arjun_1
Rahul -> Arjun_2 -> Aarav
Rahul -> Neha_2 -> Priya
Rahul -> Neha_1 -> Raj
Rahul -> Neha_1 -> Akash
```

```
Enter 1 for BFS, 2 for DFS:2
Enter start node:Rahul
Rahul -> Sneha -> Sunil -> Raj -> Neha_1 -> Akash -> Priya -> Aarav -> Neha_2 -> Arjun_2
Rahul -> Sneha -> Sunil -> Maya -> Arjun_1 -> Pooja
```